Temporal Fusion Transformers for Interpretable Multi-horizon Time Series Forecasting

https://arxiv.org/pdf/1912.09363

0. Introduction

- 시계열 예측은 다양한 산업에서 중요
- 기존 모델 한계
  - 복잡한 패턴 학습 부족
  - 해석 불가능 (black-box)
- 목표
  → 정확도 + 해석 가능성 + 다양한 입력 처리

1. Overview

- Transformer 기반 시계열 모델
- 다양한 입력을 분리 처리

입력

- static
- past observed
- known future

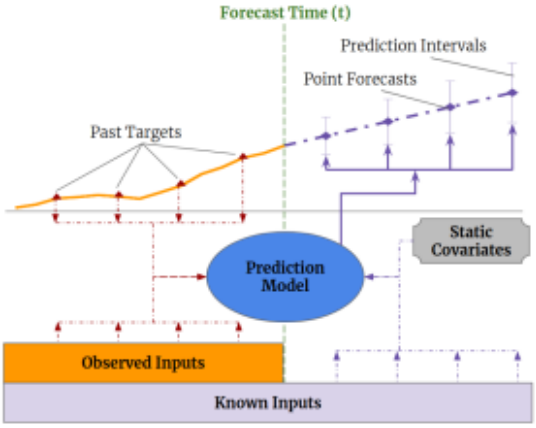
출력

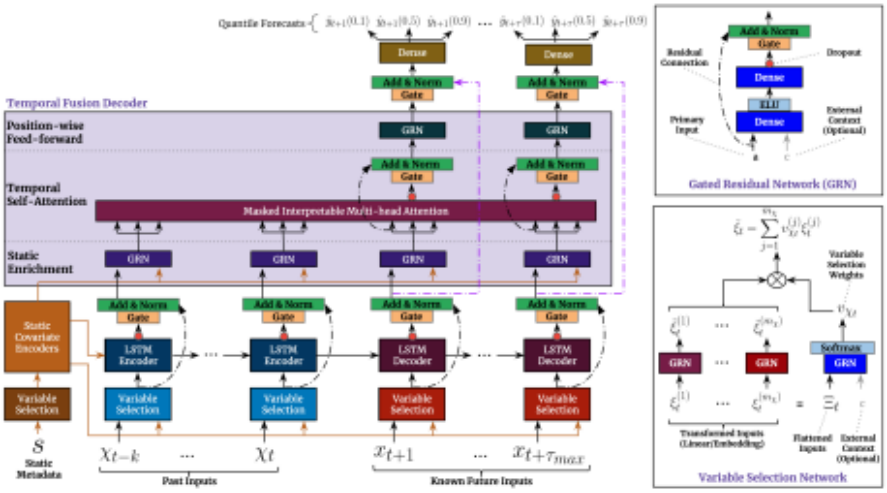
- multi-horizon forecasting

2. Challenges

- 장기 + 단기 의존성 혼합
- 다양한 변수 타입 처리
- 해석 가능성 부족
- 불필요한 변수 문제

3. Method





- Variable Selection
  → 중요한 변수만 선택
- LSTM encoder
  → 시계열 패턴 학습
- Attention
  → 시간 중요도 해석

- Gating
  → 불필요 정보 제거

4. Experiments

- 데이터
  - Electricity, Traffic, Volatility
- 비교
  - ARIMA, LSTM, DeepAR
- 지표
  - MAE, RMSE, Quantile Loss

5. Results

	Electricity	**Traffic**	**Retail**	**Vol.**
Dataset Details				
Target Type	$\mathbb{R}$	[0, 1]	$\mathbb{R}$	$\mathbb{R}$
Number of Entities	370	440	130k	41
Number of Samples	500k	500k	500k	~100k
Network Parameters				
k	168	168	90	252
τ_{max}	24	24	30	5
Dropout Rate	0.1	0.3	0.1	0.3
State Size	160	320	240	160
Number of Heads	4	4	4	1
Training Parameters				
Minibatch Size	64	128	128	64
Learning Rate	0.001	0.001	0.001	0.01
Max Gradient Norm	0.01	100	100	0.01

	ARIMA	ETS	TRMF	DeepAR	DSSM
Electricity	0.154 (*+180%*)	0.102 (*+85%*)	0.084 (*+53%*)	0.075 (*+36%*)	0.083 (*+51%*)
Traffic	0.223 (*+135%*)	0.236 (*+148%*)	0.186 (*+96%*)	0.161 (*+69%*)	0.167 (*+76%*)
	ConvTrans	**Seq2Seq**	**MQRNN**	**TFT**	
Electricity	0.059 (*+7%*)	0.067 (*+22%*)	0.077 (*+40%*)	**0.055***	
Traffic	0.122 (*+28%*)	0.105 (*+11%*)	0.117 (*+23%*)	**0.095***	

(a) P50 losses on simpler univariate datasets.

	ARIMA	ETS	TRMF	DeepAR	DSSM
Electricity	0.102 (*+278%*)	0.077 (*+185%*)	-	0.040 (*+48%*)	0.056 (*+107%*)
Traffic	0.137 (*+94%*)	0.148 (*+110%*)	-	0.099 (*+40%*)	0.113 (*+60%*)
	ConvTrans	**Seq2Seq**	**MQRNN**	**TFT**	
Electricity	0.034 (*+26%*)	0.036 (*+33%*)	0.036 (*+33%*)	**0.027***	
Traffic	0.081 (*+15%*)	0.075 (*+6%*)	0.082 (*+16%*)	**0.070***	

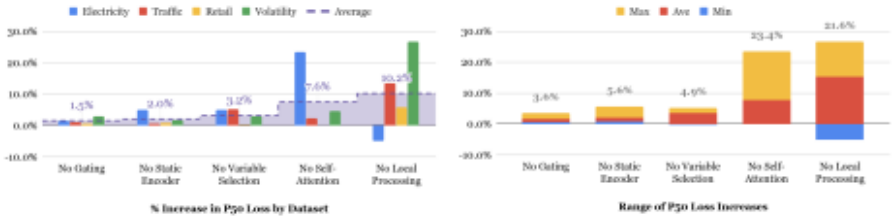
(b) P90 losses on simpler univariate datasets.

	DeepAR	CovTrans	Seq2Seq	MQRNN	TFT
Vol.	0.050 (*+28%*)	0.047 (*+20%*)	0.042 (*+7%*)	0.042 (*+7%*)	**0.039***
Retail	0.574 (*+62%*)	0.429 (*+21%*)	0.411 (*+16%*)	0.379 (*+7%*)	**0.354***

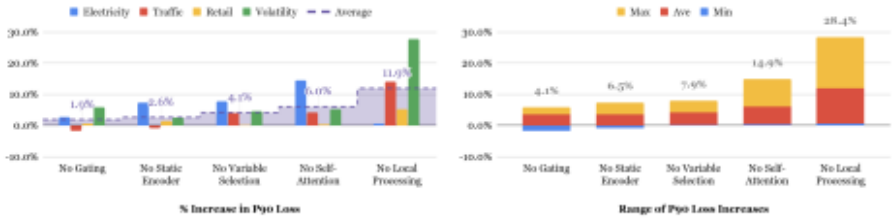
(c) P50 losses on datasets with rich static or observed inputs.

	DeepAR	CovTrans	Seq2Seq	MQRNN	TFT
Vol.	0.024 (*+21%*)	0.024 (*+22%*)	0.021 (*+8%*)	0.021 (*+9%*)	**0.020***
Retail	0.230 (*+56%*)	0.192 (*+30%*)	0.157 (*+7%*)	0.152 (*+3%*)	**0.147***

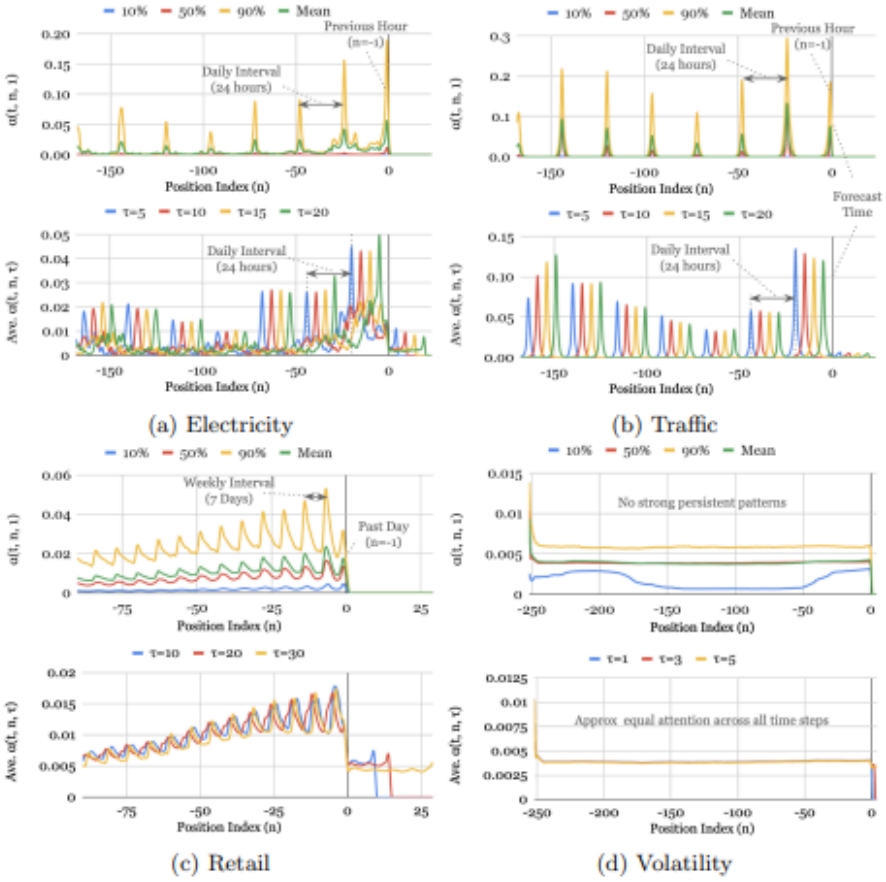
(d) P90 losses on datasets with rich static or observed inputs.



(a) Changes in P50 losses across ablation tests



(b) Changes in P90 losses across ablation tests



- 기존 모델 대비 성능 향상
- long-term forecasting에서 강점
- ablation
  - 변수 선택 제거 시 성능 감소
- 해석 가능
  - 중요한 변수 및 시점 확인 가능

6. Insight

- 해석 가능성 중요한 요소
- 변수 선택 + attention 조합 효과적
- 성능 + 해석 모두 확보
- 구조 복잡, 비용 증가